

# Aniruddh Ballal

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Engineer working across physics and production software, from solar coronal-field simulations and orbital mechanics to full-stack financial systems and modelling. Strong in end-to-end development, scientific simulation, and debugging complex systems, with a focus on understanding how they behave.

## Education

**Manipal Institute of Technology** Manipal, KA
*B.Tech, Information Technology* *Expected Jul 2026*
CGPA: 8.1

**Deeksha Integrated PU College** Yelahanka, KA
Karnataka 2nd PUC (94.2%) 2022

**Daffodils English School** Sanjaynagar, KA
ICSE 10th (92.9%) 2020

## Skills

**Languages**

TypeScript, Python, JavaScript, Rust, Kotlin, C/C++, SQL

**Frontend & Mobile**

React, Next.js, React Native, Three.js, Tailwind CSS, MUI, Redux Toolkit

**Backend & Data**

FastAPI, Node.js, Express, PostgreSQL, SQLAlchemy, MongoDB, Firebase

**Scientific & Systems**

NumPy, SciPy, XGBoost, PuLP, Astropy, Matplotlib, WASM, Web Workers, GLSL, Git, Docker, KiCad

## Experience

**Assistant Software Engineer | Indiassetz** Mar 2026–Oct 2026
Worked across React Native and Next.js/TypeScript platforms: migrated mobile applications for Android 16 KB support; built primary-sales and payment workflows in CRM; built ARCA's FastAPI/PostgreSQL investor platform with auth, document access, capital calls, settlements, and portfolio reporting.

**Data Analyst Intern | Qwixpert Consulting** Jan–Mar 2026
Built and deployed end-to-end full-stack tools, translating client data and legacy Excel workflows into scalable, high-performance solutions; diagnosed real-world problems and delivered accurate, maintainable systems.

**Frontend Web Dev Intern | Indiassetz** Jun–Jul 2025
Collaborated with design and engineering teams to enhance UI/UX with React and Material UI; implemented micro-interactions and component redesigns using clean, reusable, version-controlled code.

**Solar Physics Intern | Dept. of Physics, IISc** Jun–Aug 2024
Worked under Dr. Prantika Bhowmik on a Python project modelling and visualising solar magnetic-field evolution and cycles with spherical harmonics, SciPy, AstroPy, NumPy, and Matplotlib; derived quantitative insights into solar-cycle behaviour.

## Community

**Astronomy Club, MIT Manipal** Feb 2024 – Apr 2025
Managing Committee Member; completed coursework in AstroTech, black holes, astrobiology, and data-driven astronomy; organised club events and outreach.

**Revels PnP, MIT Manipal** Jan – Mar 2024
Led a 4-person team to paint 10+ large acrylic posters and build 3D models for MIT's annual cultural festival (25,000+ attendees).

**Manipal Runners' Club** Dec 2023 – Feb 2024
Co-organised the Manipal Marathon 2024 (8,000+ participants) and coached enrollees through a 6-week training programme.

**Junior Avionics Member | ThrustMIT** Nov 2022–Nov 2023
Built avionics for student rocketry: custom PCBs, telemetry, RF/GPS/IMU integration, microcontrollers, and control theory.

## Projects *demos & source: aniruddhballalportfolio.vercel.app*

**Sunalyse** 2025–26
Mapped the Sun's invisible coronal magnetic field into an explorable 3D scene across 181 Carrington Rotations; reconstructed curved field lines from magnetograms and cut processing time **7.6x**.

**Queen's Gambit 2.0** 2025–26
Turned chess games into a file carrier: built a reversible system that hides arbitrary files in legal moves, protects them with authenticated encryption, and makes encryption/decryption **~38x/~37x faster**.

**Credex** 2026
Built a digital trade desk for SMEs, connecting negotiation, payment, lending, and underwriting in one workflow; backed it with immutable records, cash-flow planning, and 300–900 credit scoring.

**LOANlo** 2025
Turned loan applications into a complete review workflow, from guided onboarding and document submission to underwriter decisions and administrator-led recovery, with automated risk checks and layered access controls.

**VenPay** 2025
Turned vendor credit into a trackable procurement workflow: companies request products, vendors approve them, and payments remain visible through flexible deadlines, partial settlement, and post-payment ratings.

**SatLies** 2026
Made orbital mechanics explorable: users shape an orbit and watch Earth's oblateness and lunar/solar gravity pull it away from a clean baseline in real time.